Ricardo Rossiter Barioni

AI Engineer
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Summary

Ricardo is a Lead Software Engineer with 9 years of experience developing and deploying machine learning solutions using Python, TensorFlow and PyTorch. He has worked on a wide range of projects projects involving Deep Learning, Computer Vision, Natural Language Processing, Large Language Models, and Audio/Speech Processing — with applications ranging from human pose estimation and face recognition to acoustic source classification. Over the past year, he has also been working with AWS, leveraging cloud infrastructure to build and deploy scalable ML systems. Throughout his career, he has contributed to the development of intelligent systems in both research and industry contexts, building robust models and tools that support real-world impact.

He holds both a Bachelor's and a Master's degree in Computer Science from the Universidade Federal de Pernambuco (UFPE), with academic research focused on human pose tracking and dataset creation for pose estimation. During this time, he led multiple applied research initiatives, publishing peer-reviewed papers and collaborating with industry partners. His academic journey also includes work in Virtual and Augmented Reality, combining AI with immersive technologies to enhance user experience in healthcare and the arts.

Driven by curiosity and a passion for problem-solving, Ricardo is committed to building impactful AI systems that bridge the gap between research and real-world needs. He brings a hands-on, detail-oriented mindset to his work, always striving for clarity, robustness, and innovation. For him, technology is not just a tool — it's a way to empower people, improve lives, and explore new possibilities through intelligent solutions.

Skills

Technical: Python, TensorFlow, PyTorch, AWS, OpenCV, scikit-learn, TensorFlow Lite, Librosa, Git, Docker, Jira, MongoDB, Bash, SQL, CI/CD, Computer Vision, Machine Learning, Data Science, Deep Learning, Transformers, Generative AI, LLMs, NLP, Code Review, Scrum, Kanban
Languages: Portuguese (native), English (proficient)

Experience

•Lead Computer Vision Engineer @ FCx Labs

Jan 2025 - Current
Recife, Brazil

As a Lead Computer Vision Engineer at FCx Labs, I was involved in the development and enhancement of e-commerce product catalogs to improve customer experience through high-quality visual content. My work focused on designing and deploying computer vision solutions that automated and optimized the way product data was processed, displayed, and categorized across digital platforms.

Responsibilities
• Technical leadership of a team of four Computer Vision developers, providing technical guidance and delegating tasks effectively to meet project goals.
• Collaboration with stakeholders and cross-functional business teams to define project scope, assess feasibility, and gather detailed technical requirements.
• Design and implementation of APIs and services for core computer vision tasks such as background removal, image captioning, product categorization from images, automatic image quality validation, virtual staging of products, and visual-based classification.
• Architecture of data pipelines for importing and processing products into the catalog, ensuring consistency and scalability across the system.
• Deployment of computer vision algorithms using AWS infrastructure, ensuring reliability, scalability, and performance in production environments.

Achievements and Results
• Standardized and automated the catalog image onboarding process using computer vision algorithms, reducing manual effort and errors, and bringing processing times down to just a few minutes.
• Deployed an image-based product categorization model capable of accurately classifying thousands of

products with an accuracy of approximately 90%, enhancing the speed and precision of catalog organization.
• Built a cost-effective batch image inference pipeline on AWS SageMaker, processing large datasets at a cost on the order of a few cents to a few dollars per batch.

Skills
• Python | AWS | Git | Docker | Rancher | TensorFlow | PyTorch | OpenCV | MongoDB | FastAPI | Jira

•Applied Machine Learning Scientist @ SiDi *Jan 2021 - Jan 2025*
Campinas, Brazil

As an Applied Machine Learning Scientist at SiDi, I was part of the Speech Processing team, developing machine learning and deep learning solutions to bridge the gap between academic research and real-world applications. I focused on designing and deploying models for embedded systems, targeting Samsung's mobile and IoT devices, with emphasis on efficient, production-ready solutions in audio, speech, and natural language processing.

Responsibilities
• Engagement in regular meetings with international stakeholders to present experimental results, discuss technical challenges, and align development efforts with business goals.
• End-to-end development of machine learning models, including training, validation, testing, deployment, parameter tuning, and retraining, tailored to speech and audio applications.
• Design of lightweight and optimized models for embedded deployment, focusing on size, latency, and memory efficiency using techniques such as quantization, pruning, and TensorFlow Lite.
• Direction and execution of research initiatives resulting in scientific publications and patent filings in audio and speech processing.
• Delivery of internal technical talks to promote team-wide knowledge sharing and skill development.

Achievements and Results
• Built an embedded keyword spotting system with 95% accuracy, robust to noise and reverberation, deployed in the market and used by millions of users.
• Published three peer-reviewed scientific papers on acoustic source classification, introducing novel methodologies and benchmarks.
• Co-authored 1 INPI patent.
• Developed a customizable, fully embedded keyword detection system for user-defined wake words, achieving 92% accuracy and successful deployment on commercial devices.

Skills
• Python | TensorFlow | Git | Docker | TensorFlow Lite | TFLite Micro | Jira

•Academic Researcher @ Voxar Labs *Aug 2016 - Aug 2020*
Recife, Brazil

As an Academic Researcher at Voxar Labs, I contributed to projects at the intersection of computer vision, machine learning, and virtual/augmented reality. My work combined foundational research with applied innovation, often in collaboration with industry partners to deliver research-driven solutions with real-world impact. I explored a wide range of topics, from human pose estimation to thermal-based bat tracking and immersive applications, leading to scientific publications.

Responsibilities
• Academic research in computer vision, machine learning, and related fields, contributing to theoretical and practical advancements.
• Collaboration with industry to align research objectives with business needs and deliver research as a product.
• Investigation and evaluation of state-of-the-art computer vision techniques with a focus on feasibility and real-world applicability.

Achievements and Results
• Published 8 peer-reviewed papers across diverse fields such as human pose estimation and augmented reality.

• Built a flexible framework for rapid creation of human pose estimation datasets, significantly reducing data collection and annotation time.
• Developed a virtual reality application designed to guide ballet dancers through the learning and practice of fundamental arm positions.
• Led research on 3D object reconstruction from RGB images.
• Developed and deployed a tool for visualizing bat tracking data from thermal videos, enabling new insights in ecological research.
• Researched robust face recognition methods for varied image conditions.
• Designed and validated a proof-of-concept augmented reality system to support physiotherapy rehabilitation, demonstrating its potential for patient engagement and remote therapy.
• Successfully transferred research outcomes into commercial contexts through partnerships with private companies.

Skills
• Python | Keras | PyTorch | OpenCV | C++ | Unity | Git | Docker | LaTeX

Education

•M.Sc. in Computer Science *Aug 2018 - Jul 2020*
Universidade Federal de Pernambuco Recife, Brazil

•B.Sc. in Computer Science *Apr 2014 - Jul 2018*
Universidade Federal de Pernambuco Recife, Brazil

Certifications

•Deploying Machine Learning Models in Production
DeepLearning.AI, Coursera

•Machine Learning Modeling Pipelines in Production
DeepLearning.AI, Coursera

•Probability & Statistics for Machine Learning & Data Science
DeepLearning.AI, Coursera

•Introduction to Embedded Machine Learning
Edge Impulse, Coursera

•Machine Learning Data Lifecycle in Production
DeepLearning.AI, Coursera

•Types of Conflict
University of California Irvine, Coursera

•Conflict Resolution Skills
University of California Irvine, Coursera

•Mathematics for Machine Learning: Linear Algebra
Imperial College London, Coursera

•Communication in the 21st Century Workplace
University of California Irvine, Coursera

•Effective Problem-Solving and Decision-Making
University of California Irvine, Coursera

•Work Smarter, Not Harder: Time Management for Personal & Professional Productivity
University of California Irvine, Coursera

•Digital Signal Processing 1: Basic Concepts and Algorithms
EPFL, Coursera

•Device-based Models with TensorFlow Lite
DeepLearning.AI, Coursera

•Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
DeepLearning.AI, Coursera

•Introduction to Machine Learning in Production
DeepLearning.AI, Coursera

•NLP / Sequence Models
DeepLearning.AI, Coursera

Publications

- **Improving Non-Stationary Acoustic Source Classification with Metric Learning** *Oct 2023*
  Paper at 2023 XLI Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)
- **Non-Stationarity Objective Assessment for Acoustic Source Classification** *Oct 2023*
  Paper at 2023 XLI Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)
- **A Metric Learning Based Solution for Non-Stationary Acoustic Source Classification** *Sep 2022*
  Paper at 2022 XL Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)
- **HuTrain: a Framework for Fast Creation of Real Human Pose Datasets** *Jul 2020*
  Poster at 2020 21st International Symposium on Mixed and Augmented Reality (ISMAR)
- **Songverse: a music-loop authoring tool based on Virtual Reality** *Jul 2020*
  Extended Paper at 2020 11st Journal on Interactive Systems (JIS)
- **Usability and effects of text, image and audio feedback on exercise correction during augmented reality based motor rehabilitation** *Sep 2019*
  Elsevier Computer & Graphics (C&G) Special Issue at 2019 21th Symposium on Virtual and Augmented Reality (SVR)
- **BalletVR: a Virtual Reality System for Ballet Arm Positions Training** *Aug 2019*
  Full paper at 2019 21th Symposium on Virtual and Augmented Reality (SVR)
- **Songverse: a music-loop authoring tool based on Virtual Reality** *Aug 2019*
  Full paper at 2019 21th Symposium on Virtual and Augmented Reality (SVR)
- **Human Pose Tracking from RGB Inputs** *Aug 2018*
  Full paper at 2018 20th Symposium on Virtual and Augmented Reality (SVR)
- **ARkanoidAR 2.0: Otimizações em uma solução de realidade aumentada com base em testes de usabilidade** *Aug 2018*
  Poster at 2018 26th Congresso Brasileiro de Engenharia Biomédica (CBEB)
- **ARkanoidAR: an Augmented Reality System to Guide Biomechanical Movements at Sagittal Plane** *Jun 2017*
  Full paper at 2017 19th Symposium on Virtual and Augmented Reality (SVR)